```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 1
Enter enrolment ID: 4421
Enter student name: V
Enter department: CSE
Student inserted successfully.
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 2
Enter enrolment ID to search: 4421

Student Found!
Enrolment ID: 4421
Name: V
Department: CSE
Search comparisons: 1
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 3
Enter enrolment ID to delete: 4421
Student deleted successfully.
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 4

--- Inorder Traversal ---
ID: 123 | Name: S | Department: CSE
ID: 456 | Name: V | Department: CSE
ID: 789 | Name: A | Department: CSE
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 5

--- Preorder Traversal ---
ID: 123 | Name: S | Department: CSE
ID: 456 | Name: V | Department: CSE
ID: 789 | Name: A | Department: CSE
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 6

--- Postorder Traversal ---
ID: 789 | Name: A | Department: CSE
ID: 456 | Name: V | Department: CSE
ID: 123 | Name: S | Department: CSE
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 7
10 sample records inserted successfully.
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 5

--- Preorder Traversal ---
ID: 123 | Name: S | Department: CSE
ID: 50 | Name: Aman | Department: CSE
ID: 30 | Name: Priya | Department: ECE
ID: 20 | Name: Simran | Department: CSE
ID: 10 | Name: Anjali | Department: IT
ID: 40 | Name: Karan | Department: IT
ID: 35 | Name: Vikas | Department: ECE
ID: 70 | Name: Rahul | Department: IT
ID: 60 | Name: Neha | Department: ECE
ID: 65 | Name: Pooja | Department: CSE
ID: 80 | Name: Rohit | Department: CSE
ID: 456 | Name: V | Department: CSE
ID: 789 | Name: A | Department: CSE
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 8

--- Insertion Order Comparison ---
Balanced-looking order comparisons: 10
Sorted order comparisons: 21

Balanced-looking order creates a shorter tree.
Sorted order creates a skewed tree.
```

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```
===== STUDENT RECORD SEARCH SYSTEM =====
1. Insert Student
2. Search Student
3. Delete Student
4. Inorder Traversal (Sorted Records)
5. Preorder Traversal
6. Postorder Traversal
7. Insert 10 Sample Records
8. Compare Insertion Orders
9. Exit
Enter your choice: 9
Program exited successfully.

c:\Users\vanshika\OneDrive\Desktop\DSA2proj (1)>
```

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